Arvind Yogesh Suresh Babu

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Machine Learning Engineer and MS Data Science candidate at the University of Michigan (Dec 2026) focused on scalable ML systems across forecasting, LLMs, computer vision, and distributed infrastructure. Built training/evaluation pipelines, RAG workflows, anomaly detection systems, and production inference services with PyTorch, TensorFlow, Spark, Kafka, and FastAPI; strong in rubric-driven evaluation and telemetry workflows.

Skills

Languages: Python, SQL, R, C++, JavaScript | **ML Frameworks & Methods:** PyTorch, TensorFlow, scikit-learn, XGBoost, LightGBM, LSTM/GRU/B ARIMA/SARIMA, training/benchmarking, hyperparameter tuning | **LLMs & Retrieval:** Hugging Face Transformers, RAG, embedding pipelines, vector search, prompt engineering, LLM output evaluation | **MLOps & Systems:** FastAPI, Docker, MLflow, CI/CD (Jenkins, GitHub Actions), model serving, REST APIs, schema/data-quality validation | **Data & Infrastructure:** Spark, Kafka, Databricks, dbt, Pandas, NumPy, SciPy, telemetry processing, AWS (S3, CloudFront)

Work Experience

M-City, UMTRI, Ann Arbor, MI September 2025 - Present

Research Associate

- Built Python-based model evaluation pipelines for perception systems and standardized 10+ KPIs (precision, recall, F1, latency p50/p95), improving comparison reliability by 15
- Benchmarked detection and sensor-fusion variants across edge cases, then converted failure analysis into prioritized model and pipeline improvements.
- Built reproducible validation workflows, scoring rubrics, and automated experiment scripts to reduce model-comparison turnaround.

Vestas Wind Technology, Chennai, India June 2024 - July 2025

Data Scientist Intern

- Implemented production-style forecasting pipelines (ARIMA, Holt-Winters, regression) across 200+ SKUs with temporal features and automated preprocessing, reducing manual effort by 20
- Built Kafka/Spark telemetry workflows with alerting and anomaly detection for operational monitoring, contributing to a 17
- Shipped CI-integrated, deployment-ready forecasting artifacts (Dockerized jobs, scheduled validations, KPI outputs) for reliable recurring optimization cycles.

MES section IIT Madras, Chennai, India June 2023 - Aug 2023

Research Intern

- Processed multi-channel robotic telemetry and engineered statistical and spectral features using Butterworth and notch filtering, improving signal quality by 20
- Built computer-vision tracking workflows and benchmarked Ridge, Random Forest, and Gradient Boosting models with k-fold validation, achieving under 15
- Developed reusable Python tooling for sensor integration and data acquisition to improve experiment repeatability and reduce manual effort by 20

Education

University of Michigan, Ann Arbor, MI Dec 2026
MS Data Science
Relevant Subjects: Applied Machine Learning, Natural Language Processing, Regression Analysis, Advanced DBMS, Advanced Probability and Distributions, Statistical Inference

Anna University, Chennai, India June 2024
BE Mechanical Engineering
Relevant Subjects: C Programming, Python Programming, Data Structures & Algorithms, Software Engineering, Deep Learning, Control Systems, Applied Mathematics

Certifications & Award

Microsoft: Foundations of AI and Machine Learning Infrastructure (Aug 2025, Credential ID: Q6D72J7U7V2J) |
Google: Advanced Data Analytics Specialization (Oct 2024, Credential ID: BQZ2YSKNFXVO) |
Indian Patent Office: Certificate of Registration of Design (Jul 2024, Credential ID: 404013-001)

Project Work

- **Deep Learning Forecasting System (PyTorch)**: Built an end-to-end forecasting stack with LSTM, GRU, and BiLSTM+attention using leakage-safe scaling, sliding-window datasets, Huber loss, gradient clipping, and early stopping. Implemented reusable training/evaluation workflows with RMSE/MAE/SMAPE reporting and reproducible inference artifacts.
- **MLOps-Ready Forecasting API & Feature Platform**: Engineered a FastAPI forecasting service with startup model loading, Pydantic validation, confidence intervals, and health endpoints. Built a config-driven feature engineering framework with schema validation, transformer registry, pipeline persistence, CLI automation, and unit tests.
- **Telemetry Anomaly Detection & Multivariate Forecasting**: Built a multivariate anomaly detection benchmark (LSTM autoencoder, Isolation Forest, rolling MAD, Z-score) with precision/recall/F1/ROC-AUC evaluation. Implemented multi-step forecasting (XGBoost/LightGBM) with lag, rolling, and cyclical features for high-throughput telemetry scenarios.